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CLAMPING MODULE, GRABBING DEVICE, BATTERY PRODUCTION LINE, CLAMPING METHOD, AND TRANSFER METHOD

Abstract

A clamping jaw includes a first clamping piece and a second clamping piece. A direction in which the first clamping piece and the second clamping piece are arranged opposite each other is a first direction. A first drive component is configured to drive the first clamping piece to move along the first direction. A mounting base is arranged on the second clamping piece. The second clamping piece is able to move relative to the mounting base along the first direction. The mounting base is provided with a force applying portion that is spaced away from the second clamping piece. An elastic member is in contact with the second clamping piece and the force applying portion. A second drive component is configured to drive the mounting base to move along the first direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This disclosure is a continuation of International Patent Application No. PCT/CN2023/139992, filed Dec. 19, 2023, which claims priority to Chinese Patent Application No. 202311323034.X, filed on Oct. 13, 2023, entitled “CLAMPING MODULE, GRABBING DEVICE, BATTERY PRODUCTION LINE, CLAMPING METHOD, AND TRANSFER METHOD”, each are incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] This disclosure relates to the field of battery production technologies, and in particular, to a clamping module, a grabbing device, a battery production line, a clamping method, and a transfer method.

BACKGROUND

[0003] New energy batteries are increasingly widely used in life and industry. For example, new energy vehicles equipped with batteries have been widely used. In addition, batteries are further increasingly used in the field of energy storage and the like.

[0004] In the related art, in a process of grabbing a cell (also referred to as a battery cell), a grabbing device may damage the battery cell or may be unable to stably grab the battery cell.

SUMMARY

[0005] To resolve the foregoing technical problems, this disclosure provides a clamping module, a grabbing device, a battery production line, a clamping method, and a transfer method, to reduce a possibility of battery damage while stably grabbing a battery cell.

[0006] This disclosure is implemented by using the following technical solutions.

[0007] A first aspect of embodiments of this disclosure provides a clamping module, including:

[0008] a clamping jaw, including a first clamping piece and a second clamping piece that are arranged opposite each other, where a direction in which the first clamping piece and the second clamping piece are arranged opposite each other is a first direction; [0009] first drive component, configured to drive the first clamping piece to move along the first direction; [0010] a mounting base arranged on the second clamping piece, where the second clamping piece is able to move relative to the mounting base along the first direction, and the mounting base is provided with a force applying portion that is spaced away from the second clamping piece; [0011] an elastic member in contact with the second clamping piece and the force applying portion, to enable the force applying portion to apply an acting force to the second clamping piece through the elastic member; and [0012] a second drive component, configured to drive the mounting base to move along the first direction, so that a direction of the acting force applied by the elastic member to the second clamping piece is toward the first clamping piece.

[0013] In this embodiment of this disclosure, the first drive component and the second drive component independently drive the first clamping piece and the mounting base, so that the

clamping jaw can stably clamp a battery cell while minimizing damage to the battery cell.

[0014] In an embodiment, the force applying portion is located on a side of the second clamping piece that is away from the first clamping piece along the first direction.

[0015] In this embodiment of this disclosure, the elastic member is at least partially located on the side of the second clamping piece that is away from the first clamping piece along the first direction, so that space originally formed between the mounting base and the side of the second clamping piece that is away from the first clamping piece along the first direction is fully utilized.

[0016] In an embodiment, the mounting base is further provided with a position limiting portion, the position limiting portion is located on a side of the force applying portion that faces the first clamping piece along the first direction, the second clamping piece is located between the position limiting portion and the force applying portion along the first direction, and the second clamping piece abuts against the position limiting portion under the action of an elastic force of the elastic member.

[0017] In this embodiment of this disclosure, the second clamping piece is pressed against the position limiting portion by the elastic force of the compressed elastic member, so that shaking of the second clamping piece between the position limiting portion and the elastic member is reduced.

[0018] In an embodiment, the elastic member is located on a side of the second clamping piece that is away from the position limiting portion, and the elastic member is located between the second clamping piece and the force applying portion.

[0019] In this embodiment of this disclosure, the elastic member is fully located in space originally formed between the force applying portion and a side of the second clamping piece that is away from the first clamping piece, and the space between the force applying portion and the side of the second clamping piece that is away from the first clamping piece is fully utilized to accommodate the elastic member, so that space utilization is improved.

[0020] A second aspect of embodiments of this disclosure provides a grabbing device, including:

[0021] a frame; [0022] clamping assemblies arranged on the frame, where there are at least two clamping assemblies, each clamping assembly includes a base and the clamping module according to any one of the foregoing implementations, at least two clamping assemblies are arranged along a second direction, the second direction intersects with the first direction, the first drive component, the second drive component, the first clamping piece, and the mounting base are all arranged on the base, the first drive component drives the first clamping piece to move relative to the base, the second drive component drives the mounting base to move relative to the base, each clamping assembly has at least one clamping module, and each clamping module of each clamping assembly and a corresponding clamping module of another corresponding clamping assembly are arranged along the second direction; and [0023] a drive assembly arranged on the frame, where the drive assembly is configured to drive one clamping assembly to move relative to another clamping assembly along the second direction.

[0024] In this embodiment of this disclosure, the drive assembly drives one clamping assembly to move relative to another clamping assembly, so that a distance between each clamping module of each clamping assembly and a corresponding clamping module of another corresponding clamping assembly along the second direction can be adjusted, to clamp battery cells at different spacings on the same conveying apparatus.

[0025] In an embodiment, each clamping assembly has a plurality of clamping modules, and clamping jaws of the plurality of clamping modules are arranged at spacings along the first direction.

[0026] In this embodiment of this disclosure, the plurality of clamping modules may jointly grab battery cells on a plurality of conveying apparatuses, to improve efficiency of grabbing battery cells.

[0027] In an embodiment, each clamping assembly has two clamping modules, two second clamping pieces corresponding to the two clamping modules are located between two first

clamping pieces along the first direction, two corresponding mounting bases are located between the two first clamping pieces along the first direction, and the second drive component is located on a side of the mounting base along the second direction.

[0028] In this embodiment of this disclosure, two corresponding mounting bases are located between two first clamping pieces along the first direction, the second drive component is located on a side of the mounting base along the second direction, and a spacing between the two mounting bases at limiting positions away from the first clamping pieces along the first direction can be arranged to be small, to help reduce an overall size of the grabbing device along the first direction.

[0029] In an embodiment, a second drive component of a clamping module of each clamping assembly is located on a side of a corresponding mounting base that is away from another corresponding clamping assembly along the second direction, and two second drive components corresponding to two clamping modules are located between two first clamping pieces along the first direction.

[0030] In this embodiment of this disclosure, a second drive component of a clamping module of each clamping assembly is located on a side of a corresponding mounting base that is away from another corresponding clamping assembly along the second direction, and the two clamping assemblies can approach each other as close as possible along the second direction. This helps grab battery cells at different spacings on the same conveying apparatus within a larger spacing range.

[0031] In an embodiment, a drive portion is formed at an end of each second drive component that faces a corresponding first clamping piece, the drive portion is connected to an end of a corresponding mounting base that faces a corresponding first clamping piece, and two second drive components are located between two drive portions along the first direction.

[0032] In this embodiment of this disclosure, because the drive portion is connected to an end of a corresponding mounting base that faces a corresponding first clamping piece, two mounting bases with large sizes along the first direction can be arranged in a region between two drive portions along the first direction, so that the region between the two drive portions spaced apart along the first direction can be fully utilized, to improve space utilization.

[0033] In an embodiment, the clamping assembly further includes a stop mechanism, the stop mechanism is located between two mounting bases along the first direction, and the stop mechanism is able to abut against the two mounting bases along the first direction to prevent the two mounting bases from approaching each other.

[0034] In this embodiment of this disclosure, the stop mechanism abuts against the two mounting bases, so that a possibility of interference between the two mounting bases **23** due to approaching is reduced.

[0035] In an embodiment, the stop mechanism includes: [0036] a supporting base connected to the base; [0037] a stop rod passing through the supporting base, where each mounting base is correspondingly provided with the stop rod, and the stop rod is configured to abut against a corresponding mounting base; and [0038] a locking piece in threaded connection to a corresponding stop rod, where each stop rod is correspondingly provided with two locking pieces, one locking piece is located on a side of the supporting base along the first direction, and the other locking piece is located on the other side of the supporting base along the first direction.

[0039] In this embodiment of this disclosure, a stop rod corresponding to each mounting base passes through the supporting base, and two side locking pieces corresponding to each stop rod are in threaded connection to the stop rod, so that the stop rod is fastened to the supporting base. When one stop rod is damaged, the corresponding stop rod may be removed and replaced with a new stop rod, without replacing the entire stop mechanism. This helps reduce maintenance costs.

[0040] In an embodiment, a base of one clamping assembly is connected to the frame, and a base of another clamping assembly is movably arranged on the frame.

[0041] In this embodiment of this disclosure, because a base of one clamping assembly is connected to the frame, the corresponding clamping assembly does not move relative to the base.

This helps simplify an assembly structure between the corresponding clamping assembly and the base. One of two clamping assemblies moves relative to the base, and the other clamping assembly does not move relative to the base, so that one clamping assembly can move relative to the other clamping assembly, and an assembly structure between the clamping assemblies and the base can also be simplified.

[0042] In an embodiment, the base includes: [0043] a main base, where the second drive component, the first clamping piece, and the mounting base are all arranged on the main base; and [0044] a position limiting base connected to the main base, where both the position limiting base and the first drive component are located on a side of the first clamping piece that is away from the second clamping piece along the first direction, the first drive component is mounted on the position limiting base, and the first drive component is able to drive the first clamping piece to move to abut against the position limiting base.

[0045] In this embodiment of this disclosure, the position limiting base is located at an end of the first clamping piece that is away from the second clamping piece, and the position limiting base can abut against the first clamping piece for position limiting, to prevent the first clamping piece from excessively moving away from the second clamping piece by a long distance. The first drive component is mounted on the position limiting base located on the side of the first clamping piece that is away from the second clamping piece. This facilitates mounting of the first drive component. In addition, the first drive component mounted on the position limiting base is located on the side of the first clamping piece that is away from the second clamping piece, so that the first drive component can smoothly push the first clamping piece to move along the first direction.

[0046] A third aspect of embodiments of this disclosure provides a battery production line, including: [0047] a conveying apparatus, configured to convey a battery cell; [0048] the grabbing device according to any one of the foregoing implementations, configured to grab the battery cell conveyed by the conveying apparatus; [0049] a stacking table, configured to receive the battery cell grabbed by the grabbing device from the conveying apparatus, to pre-stack the battery cell; and [0050] a transfer device, configured to drive the grabbing device to reciprocate between the conveying apparatus and the stacking table.

[0051] In this embodiment of this disclosure, the first drive component and the second drive component independently perform driving, so that the grabbing device can stably grab the battery cell on the conveying apparatus to the stacking table for stacking while minimizing damage to the battery cell.

[0052] A fourth aspect of embodiments of this disclosure provides a clamping method, including: [0053] driving, by using a first drive component, a first clamping piece to move to get in contact with a battery cell; [0054] driving, by using a second drive component, a mounting base to move, so that a second clamping piece moves to get in contact with the battery cell; [0055] when the battery cell located between the first clamping piece and the second clamping piece is in contact with the first clamping piece and the second clamping piece, stopping driving, by using the first drive component, the first clamping piece to move; and [0056] driving, by using the second drive component, the mounting base to move to a preset clamping position toward the first clamping piece that stops moving.

[0057] In this embodiment of this disclosure, the first drive component and the second drive component independently perform driving, so that the battery cell can be stably grabbed with operations on the battery cell minimized.

[0058] A fifth aspect of embodiments of this disclosure provides a transfer method, including: [0059] driving, by using a transfer device, a grabbing device to move to a to-be-grabbed material position on a conveying apparatus, so that a battery cell is located between a first clamping piece and a corresponding second clamping piece; [0060] driving, by using a first drive component, the first clamping piece to move to get in contact with the battery cell; [0061] driving, by using a second drive component, a mounting base to move, so that the second clamping piece moves to get

in contact with the battery cell; [0062] when the battery cell located between the first clamping piece and the second clamping piece is in contact with the first clamping piece and the second clamping piece, stopping driving, by using the first drive component, the first clamping piece to move; [0063] driving, by using the second drive component, the mounting base to move, so that the second clamping piece moves to a preset clamping position toward the first clamping piece that stops moving; and [0064] transferring, by using the transfer device, the battery cell grabbed by the grabbing device to a stacking table for stacking.

[0065] In this embodiment of this disclosure, the first drive component and the second drive component independently perform driving, so that the grabbing device can stably grab the battery cell on the conveying apparatus to the stacking table for stacking while minimizing damage to the battery cell.

[0066] In an embodiment, the transfer method further includes: [0067] driving, by using a drive assembly, one of two clamping assemblies to move relative to the other clamping assembly, so that a spacing between two clamping jaws arranged along a second direction is suitable for grabbing assembled battery cells on the conveying apparatus.

[0068] In this embodiment of this disclosure, the drive assembly drives the clamping assembly to move, to adapt to battery cells with different specifications on the conveying apparatus.

[0069] In an embodiment, there are two conveying apparatuses, the two conveying apparatuses are arranged in parallel, each clamping assembly has two clamping modules, the two clamping modules of each clamping assembly are in a one-to-one correspondence to the conveying apparatuses, and a pole of a battery cell on each conveying apparatus faces the other corresponding conveying apparatus; and [0070] the transferring, by using the transfer device, the battery cell grabbed by the grabbing device to a stacking table for stacking includes: [0071] driving, by using the transfer device, the grabbing device to move away from the conveying apparatus, so that the grabbing device is located at a position able to prevent the grabbing device from interfering with the conveying apparatus during rotation; [0072] driving, by using the transfer device, the grabbing device to rotate, so that a pole of the battery cell grabbed by the grabbing device is arranged along a vertical direction; [0073] performing stacking: driving, by using the transfer device, the grabbing device to move, so that the grabbing device transfers the grabbed battery cell whose pole faces upward to the stacking table for stacking and release; and [0074] when battery cells grabbed by the grabbing device are not all released, driving, by using the transfer device, the grabbing device that has released a battery cell to rotate in an avoidance region, so that a pole of a battery cell grabbed by the grabbing device faces upward for performing the stacking step.

[0075] In this embodiment of this disclosure, stacking is performed twice through rotation of the grabbing device, so that poles of all battery cells placed on the stacking table can be arranged upward.

[0076] In an embodiment, the transferring, by using the transfer device, the battery cell grabbed by the grabbing device to a stacking table for stacking further includes: [0077] when the battery cell grabbed by the grabbing device is single-row-assembled battery cells, driving, by using the drive assembly, one of two clamping assemblies to move relative to the other clamping assembly, so that a spacing between two clamping jaws arranged along the second direction is suitable for placing the single-row-assembled battery cells to a corresponding position on the stacking table.

[0078] In this embodiment of this disclosure, the drive assembly drives the clamping assembly to move, so that a spacing between single-row-assembled battery cells grabbed by the grabbing device can adapt to a placement position, on the stacking table, at which the single-row-assembled battery cells are to be stacked.

Effects of Invention

[0079] In a clamping module in the embodiments of this disclosure, before a clamping jaw clamps a battery cell, the battery cell is located between a first clamping piece and a second clamping piece along a first direction, and the battery cell is spaced away from the first clamping piece and the

second clamping piece by a specific distance along the first direction. In this way, in a process in which the clamping jaw moves downward to a position at which the battery cell can be grabbed, a possibility that the first clamping piece and the second clamping piece press against the battery cell and cause damage to the battery cell can be reduced. Because the battery cell is spaced away from the first clamping piece and the second clamping piece by a specific distance along the first direction, a first drive component needs to drive the first clamping piece to move along the first direction to a position for contact with the battery cell, and a second drive component needs to drive a mounting base to move along the first direction to enable the second clamping piece to move to a position for contact with the battery cell. When the battery cell is in contact with the first clamping piece and the second clamping piece, the first drive component stops driving the first clamping piece to move along the first direction, and the second drive component drives the mounting base to move toward the first clamping piece, so that a force applying portion of the mounting base gradually increases, through an elastic member, an acting force applied to the second clamping piece. A clamping force applied by the first clamping piece and the second clamping piece to clamp the battery cell is gradually increased to a required clamping force through buffering and adjustment by the elastic member, to reduce a possibility that the clamping force applied by the first clamping piece and the second clamping piece to the battery cell is excessively large and causes damage to the battery cell. In addition, the second drive component drives the mounting base to move to gradually increase the clamping force, so that the battery cell can be stably clamped. In a process in which the second drive component drives the mounting base to move toward the first clamping piece to enable the force applying portion of the mounting base to gradually increase, through the elastic member, the acting force applied to the second clamping piece, because the first drive component drives the first clamping piece to move and no corresponding elastic member is arranged between the first drive component and the first clamping piece, the first drive component rigidly drives the first clamping piece, and a position to which the first drive component performs driving determines a position of the first clamping piece. This can reduce shaking of the battery cell clamped between the first clamping piece and the second clamping piece, and help stably clamp the battery cell.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0080] Persons of ordinary skill in the art can clearly understand various other advantages and benefits by reading the detailed description of the preferred embodiments below. The accompanying drawings are merely intended to illustrate the preferred embodiments and are not intended to limit this disclosure. Moreover, same reference signs are used to represent same components throughout the accompanying drawings. In the accompanying drawings:

[0081] FIG. 1 is a schematic diagram of two conveying apparatuses arranged in parallel according to an embodiment of this disclosure;

[0082] FIG. 2 is a first schematic diagram of a stacking status of battery cells on a stacking table according to an embodiment of this disclosure, where this figure shows a stacking status of single-row-assembled battery cells on the stacking table;

[0083] FIG. 3 is a second schematic diagram of a stacking status of battery cells on a stacking table according to an embodiment of this disclosure, where this figure shows a stacking status of double-row-assembled battery cells on the stacking table;

[0084] FIG. 4 is a first schematic structural diagram of a grabbing device according to an embodiment of this disclosure, where this figure shows a drive assembly;

[0085] FIG. 5 is a second schematic structural diagram of a grabbing device according to an embodiment of this disclosure, where this figure shows no drive assembly;

[0086] FIG. 6 is an enlarged view of a position A in FIG. 5;
[0087] FIG. 7 is an exploded view of a grabbing device according to an embodiment of this disclosure;
[0088] FIG. 8 is a schematic structural diagram of a clamping assembly according to an embodiment of this disclosure;
[0089] FIG. 9 is an assembly diagram of a mounting base, an elastic member, and a second clamping piece according to an embodiment of this disclosure;
[0090] FIG. 10 is a schematic structural diagram of a stop mechanism according to an embodiment of this disclosure;
[0091] FIG. 11 is a flowchart of a clamping method according to an embodiment of this disclosure;
[0092] FIG. 12 is a first flowchart of a transfer method according to an embodiment of this disclosure;
[0093] FIG. 13 is a second flowchart of a transfer method according to an embodiment of this disclosure; and
[0094] FIG. 14 is a third flowchart of a transfer method according to an embodiment of this disclosure.

[0095] Reference signs are described as follows: [0096] **100**: frame; **200**: clamping assembly; **1**: base; **11**: main base; **12**: position limiting base; **2**: clamping module; **21**: clamping jaw; **211**: first clamping piece; **212**: second clamping piece; **22**: first drive component; **23**: mounting base; **231**: force applying portion; **232**: position limiting portion; **24**: elastic member; **25**: second drive component; **251**: drive portion; **3**: stop mechanism; **31**: supporting base; **32**: stop rod; **33**: locking piece; **300**: drive assembly; **400**: conveying apparatus; **500**: stacking table; and **600**: battery cell.

DESCRIPTION OF EMBODIMENTS

[0097] Embodiments of technical solutions of this disclosure are described in detail below with reference to the accompanying drawings. The following embodiments are merely intended for a clearer description of the technical solutions of this disclosure and therefore are used as just examples which do not constitute any limitations on the protection scope of this disclosure.

[0098] Unless otherwise defined, all technical and scientific terms used in this specification shall have the same meanings as commonly understood by persons skilled in the art to which this disclosure belongs. The terms used in this specification are merely intended to describe the specific embodiments rather than to limit this disclosure. The terms “include”, “have”, and any other variations thereof in the embodiments of this disclosure are intended to cover non-exclusive inclusions.

[0099] In the description of the embodiments of this disclosure, the technical terms “first”, “second”, “third”, and the like are merely intended to distinguish between different objects, and shall not be understood as any indication or implication of relative importance or any implicit indication of the number, sequence or primary-secondary relationship of the technical features indicated. In the description of the embodiments of this disclosure, “a plurality of” means two or more, unless otherwise explicitly and specifically defined.

[0100] An “embodiment” mentioned in this specification indicates that a particular feature, structure, or characteristic described with reference to the embodiment may be included in at least one embodiment of this disclosure. The term appearing in various places in this specification does not necessarily refer to the same embodiment or an independent or alternative embodiment that is exclusive of other embodiments. Persons skilled in the art explicitly and implicitly understand that the embodiments described herein may be combined with other embodiments.

[0101] In the description of the embodiments of this disclosure, the term “and/or” is only an associative relationship for describing associated objects, indicating that three relationships may be present. For example, A and/or B may indicate the following three cases: presence of only A, presence of both A and B, and presence of only B. In addition, the character “/” in this specification usually indicates an “or” relationship between the associated objects.

[0102] In the description of the embodiments of this disclosure, unless otherwise specified and defined explicitly, the technical terms “mounting”, “connection”, “join”, “fastening”, and the like should be understood in their general senses. For example, they may refer to a fixed connection, a detachable connection, or an integral connection, may refer to a mechanical connection or an electrical connection, and may refer to a direct connection, an indirect connection via an intermediate medium, an internal communication between two elements, or an interaction between two elements. Persons of ordinary skills in the art can understand specific meanings of these terms in this disclosure as appropriate to specific situations.

[0103] In the description of the embodiments of this disclosure, unless otherwise specified and defined explicitly, the technical term “contact” should be understood in its general sense.

[0104] It may refer to direct contact, contact via an intermediate medium layer, contact with no interaction between two elements that are in contact, or contact with an interaction between two elements that are in contact.

[0105] In the description of the embodiments of this disclosure, the orientations or positional relationships indicated by “above”, “below”, “top”, and “bottom” are based on the orientations or positional relationships shown in FIG. 1 to FIG. 3. It can be understood that these orientation terms are merely for ease and brevity of description of this disclosure rather than indicating or implying that the means or components mentioned must have specific orientations or must be constructed or manipulated according to specific orientations, and therefore shall not be construed as any limitation on this disclosure.

[0106] In the embodiments of this disclosure, as shown in FIG. 5, a first direction is a direction indicated by an arrow R1 in the figure.

[0107] In the embodiments of this disclosure, as shown in FIG. 5, a second direction is a direction indicated by an arrow R2 in the figure.

[0108] In the embodiments of this disclosure, as shown in FIG. 1 to FIG. 3, a vertical direction is a direction indicated by an arrow R3 in the figures.

[0109] In the embodiments of this disclosure, as shown in FIG. 1, a conveying direction of a conveying apparatus is a direction indicated by an arrow R4 in the figure.

[0110] As a part of a creative concept of this disclosure, before the embodiments of this disclosure are described, a reason why a grabbing device may damage a battery cell or be unable to stably grab the battery cell in a process of grabbing the battery cell in the related art is analyzed, to obtain the technical solutions in the embodiments of the disclosure through appropriate analysis.

[0111] In the related art, a battery cell is clamped between a first clamping piece and a second clamping piece by the first clamping piece and the second clamping piece. When clamping forces applied to the first clamping piece and the second clamping piece are not buffered or adjusted by an elastic member, a small change in a distance between the first clamping piece and the second clamping piece may cause large fluctuation in the clamping forces, and it is difficult to accurately control clamping forces applied by the first clamping piece and the second clamping piece to the battery cell. Consequently, the battery cell may be damaged, or the battery cell may not be tightly clamped. When the clamping forces applied to the first clamping piece and the second clamping piece are buffered and adjusted by the elastic member, both the first clamping piece and the second clamping piece are affected by an elastic force of the elastic member. Consequently, positions of the first clamping piece and the second clamping piece are uncertain. This may cause shaking of the battery cell clamped by the first clamping piece and the second clamping piece, and it is difficult for the first clamping piece and the second clamping piece to stably grab the battery cell. In addition, there is a linkage relationship between the first clamping piece and the second clamping piece. The first clamping piece and the second clamping piece are not independently driven, and a position of the battery cell on a conveying apparatus is fixed after the conveying apparatus conveys the battery cell to a corresponding to-be-grabbed position. In a case that an acting force applied to the first clamping piece is not buffered or adjusted by the elastic member, an

acting force applied to the second clamping piece can be buffered and adjusted by the elastic member, and both the first clamping piece and the second clamping piece are in contact with the battery cell, because the first clamping piece and the second clamping piece are not independently driven, the battery cell at the fixed position on the conveying apparatus may be damaged due to extrusion by the first clamping piece.

[0112] In the embodiments of the disclosure, a first clamping piece and a second clamping piece are independently driven, and a clamping force applied to the second clamping piece is buffered and adjusted by an elastic member. In a process of clamping a battery cell, a first drive component drives the first clamping piece to get in contact with the battery cell, and a second drive component drives a mounting base to move to enable the second clamping piece to get in contact with the battery cell. When the battery cell is in contact with the corresponding first clamping piece and the corresponding second clamping piece, the first drive component stops driving the first clamping piece to move, to prevent the battery cell from being damaged by a clamping force applied by the first drive component to the first clamping piece without buffering or adjustment, and the second drive component drives the mounting base to move. A force applying portion of the moving mounting base applies a corresponding clamping force to the second clamping piece by compressing the elastic member. A degree of compressing the elastic member to apply the clamping force is adjustable, and a clamping force applied to the first clamping piece is not buffered or adjusted by the elastic member, so that a position of the first clamping piece is certain. This can reduce a degree of shaking of the battery cell, and help stably grab the battery cell.

[0113] The solutions in the embodiments of this disclosure may be applied to but are limited to grabbing of a battery cell **600**, and are also applicable to grabbing of other easily damaged objects.

[0114] An electric apparatus is an apparatus that uses electricity as energy and consumes electricity to implement a corresponding function. For example, the electric apparatus may be but is not limited to a mobile phone, a tablet computer, a notebook computer, an electric toy, an electric tool, an electric bicycle, an electric vehicle, a ship, a spacecraft, and the like. The electric toy may be a fixed or mobile electric toy, for example, a game console, an electric toy car, an electric toy ship, and an electric toy airplane. The spacecraft may include an airplane, a rocket, a space shuttle, a spaceship, and the like.

[0115] An example in which an electric apparatus according to an embodiment of this disclosure is a vehicle is used for description.

[0116] A vehicle provided in an embodiment of this disclosure may be an oil-fueled vehicle, a gas-powered vehicle, or a new energy vehicle. The new energy vehicle may be a battery electric vehicle, a hybrid electric vehicle, an extended-range electric vehicle, or the like. The vehicle is provided with a battery pack inside, and the battery pack may be arranged at the bottom, front, or rear of the vehicle. The battery pack may be configured to supply power to the vehicle. For example, the battery pack may be used as an operational power supply for the vehicle. The vehicle may further include a controller and a motor. The controller may be configured to control the battery pack to supply power to the motor. For example, the battery pack may be configured to meet power requirements of start, navigation, and driving of the vehicle.

[0117] In some embodiments of this disclosure, the battery pack can be used as not only the operational power supply for the vehicle but also a driving power supply for the vehicle, replacing all or a part of fossil fuel or natural gas to provide driving traction for the vehicle. In some embodiments, the battery pack can also be used for energy storage devices.

[0118] The battery pack includes one or more battery cells **600**.

[0119] There may be a plurality of battery cells **600**. The plurality of battery cells **600** may be connected in series, parallel, or series-parallel, where being connected in series-parallel means a combination of series and parallel connections of the plurality of battery cells **600**. The plurality of battery cells **600** may be directly connected in series, parallel, or series-parallel, and then an entirety of the plurality of battery cells **600** is placed in a box. Certainly, the plurality of battery

cells **600** may alternatively be first connected in series, parallel, or series-parallel to form a battery module, and then a plurality of battery modules are connected in series, parallel, or series-parallel to form an entirety, and the entirety formed by the plurality of battery modules connected in series, parallel, or series-parallel is placed in a box. The battery pack may further include other structures. For example, the battery pack may further include a busbar configured to implement electrical connection between the plurality of battery cells **600**.

[0120] The battery cell **600** is a basic unit that can implement mutual conversion between chemical energy and electric energy.

[0121] In the embodiments of this disclosure, the battery cell **600** may be a secondary battery. The secondary battery is a battery cell **600** that can be charged after being discharged to activate active materials for continuous use.

[0122] In the embodiments of this disclosure, the battery cell **600** may be a lithium-ion battery, a sodium-ion battery, a sodium-lithium-ion battery, a lithium metal battery, a sodium metal battery, a lithium-sulfur battery, a magnesium-ion battery, a nickel-metal hydride battery, a nickel-cadmium battery, a lead battery, or the like. This is not limited in the embodiments of this disclosure.

[0123] For a battery production line in the embodiments of this disclosure, refer to FIG. 1 to FIG. 4. The battery production line includes a conveying apparatus **400**, a grabbing device, a stacking table **500**, and a transfer device. The conveying apparatus **400** is configured to convey a battery cell **600**. The grabbing device is configured to grab the battery cell **600** conveyed by the conveying apparatus **400**. The stacking table **500** is configured to receive the battery cell **600** grabbed by the grabbing device from the conveying apparatus **400**, to pre-stack the battery cell **600**. The transfer device is configured to drive the grabbing device to reciprocate between the conveying apparatus **400** and the stacking table **500**.

[0124] For the grabbing device in the embodiments of this disclosure, refer to FIG. 4, FIG. 5, and FIG. 7. The grabbing device includes a frame **100**, clamping assemblies **200**, and a drive assembly **300**. The clamping assemblies **200** are arranged on the frame **100**. There are at least two clamping assemblies **200**. Each clamping assembly **200** includes a base **1** and a clamping module **2**. At least two clamping assemblies **200** are arranged along a second direction. The second direction intersects with a first direction. The clamping module **2** is arranged on the base **1**. Each clamping assembly **200** has at least one clamping module **2**. Each clamping module **2** of each clamping assembly **200** and a corresponding clamping module **2** of another corresponding clamping assembly **200** are arranged along the second direction. The drive assembly **300** is arranged on the frame **100**. The drive assembly **300** is configured to drive one clamping assembly **200** to move relative to another clamping assembly **200** along the second direction.

[0125] In the embodiments of this disclosure, the drive assembly **300** drives one clamping assembly **200** to move relative to another clamping assembly **200**, so that a distance between each clamping module **2** of each clamping assembly **200** and a corresponding clamping module **2** of another corresponding clamping assembly **200** along the second direction can be adjusted, to clamp battery cells **600** at different spacings on the same conveying apparatus **400**.

[0126] For example, the grabbing device may be configured to grab assembled battery cells **600** on the conveying apparatus **400**.

[0127] For example, the assembled battery cells **600** mainly include two battery cells **600**. For example, the grabbing device is configured to grab single-row-assembled battery cells **600** on the conveying apparatus **400**, or the grabbing device is configured to grab double-row-assembled battery cells **600** on the conveying apparatus **400**.

[0128] For the clamping module **2** in the embodiments of this disclosure, refer to FIG. 5, FIG. 8, and FIG. 9. The clamping module **2** includes a clamping jaw **21**, a first drive component **22**, a mounting base **23**, an elastic member **24**, and a second drive component **25**. The clamping jaw **21** includes a first clamping piece **211** and a second clamping piece **212** that are arranged opposite each other. A direction in which the first clamping piece **211** and the second clamping piece **212** are

arranged opposite each other is the first direction. The first drive component **22** is configured to drive the first clamping piece **211** to move along the first direction. The mounting base **23** is arranged on the second clamping piece **212**. The second clamping piece **212** is able to move relative to the mounting base **23** along the first direction. The mounting base **23** is provided with a force applying portion **231** that is spaced away from the second clamping piece **212**. The elastic member **24** is in contact with the second clamping piece **212** and the force applying portion **231**, to enable the force applying portion **231** to apply an acting force to the second clamping piece **212** through the elastic member **24**. The second drive component **25** is configured to drive the mounting base **23** to move along the first direction, so that a direction of an acting force applied by the elastic member **24** to the second clamping piece **212** is toward the first clamping piece **211**.

[0129] The first drive component **22** drives the first clamping piece **211** to move, and the second drive component **25** drives the mounting base **23** to move, to enable the second clamping piece **212** to move correspondingly. The first drive component **22** and the second drive component **25** may independently perform driving, so that the first clamping piece **211** and the second clamping piece **212** independently move correspondingly.

[0130] For example, according to an actual requirement, the first drive component **22** may drive the first clamping piece **211** to move, but the second drive component **25** does not drive the mounting base **23** to move.

[0131] For example, according to an actual requirement, the second drive component **25** needs to drive the second clamping piece **212** to move, but the first drive component **22** does not drive the first clamping piece **211** to move.

[0132] For example, the elastic member **24** may be a spring.

[0133] For example, as shown in FIG. 4, FIG. 5, and FIG. 7, the first drive component **22**, the second drive component **25**, the first clamping piece **211**, and the mounting base **23** are all arranged on the base **1**, the first drive component **22** drives the first clamping piece **211** to move relative to the base **1**, and the second drive component **25** drives the mounting base **23** to move relative to the base **1**.

[0134] In the embodiments of this disclosure, before the clamping jaw **21** clamps the battery cell **600**, the battery cell **600** is located between the first clamping piece **211** and the second clamping piece **212** along the first direction, and the battery cell **600** is spaced away from the first clamping piece **211** and the second clamping piece **212** by a specific distance along the first direction. In this way, in a process in which the clamping jaw **21** moves downward to a position at which the battery cell **600** can be grabbed, a possibility that the first clamping piece **211** and the second clamping piece **212** press against the battery cell **600** and cause damage to the battery cell **600** can be reduced. Because the battery cell **600** is spaced away from the first clamping piece **211** and the second clamping piece **212** by a specific distance along the first direction, the first drive component **22** needs to drive the first clamping piece **211** to move along the first direction to a position for contact with the battery cell **600**, and the second drive component **25** needs to drive the mounting base **23** to move along the first direction to enable the second clamping piece **212** to move to a position for contact with the battery cell **600**. When the battery cell **600** is in contact with the first clamping piece **211** and the second clamping piece **212**, the first drive component **22** stops driving the first clamping piece **211** to move along the first direction, and the second drive component **25** drives the mounting base **23** to move toward the first clamping piece **211**, so that the force applying portion **231** of the mounting base **23** gradually increases, through the elastic member **24**, an acting force applied to the second clamping piece **212**. A clamping force applied by the first clamping piece **211** and the second clamping piece **212** to clamp the battery cell **600** is gradually increased to a required clamping force through buffering and adjustment by the elastic member **24**, to reduce a possibility that the clamping force applied by the first clamping piece **211** and the second clamping piece **212** to the battery cell **600** is excessively large and causes damage to the battery cell **600**. In addition, the second drive component **25** drives the mounting base **23** to move to gradually increase

the clamping force, so that the battery cell **600** can be stably clamped. In a process in which the second drive component **25** drives the mounting base **23** to move toward the first clamping piece **211** to enable the force applying portion **231** of the mounting base **23** to gradually increase, through the elastic member **24**, the acting force applied to the second clamping piece **212**, because the first drive component **22** drives the first clamping piece **211** to move and no corresponding elastic member **24** is arranged between the first drive component **22** and the first clamping piece **211**, the first drive component **22** rigidly drives the first clamping piece **211**, and a position to which the first drive component **22** performs driving determines a position of the first clamping piece **211**. This can reduce shaking of the battery cell **600** clamped between the first clamping piece **211** and the second clamping piece **212**, and help stably clamp the battery cell **600**.

[0135] In an embodiment, as shown in FIG. 5, FIG. 6, FIG. 8, and FIG. 9, the force applying portion **231** is located on a side of the second clamping piece **212** that is away from the first clamping piece **211** along the first direction.

[0136] In this embodiment of this disclosure, when the second clamping piece **212** is in contact with the battery cell **600**, the second drive component **25** still needs to drive the mounting base **23** to move toward the first clamping piece **211** relative to the second clamping piece **212**, to increase an amount of compression of the elastic member **24** and therefore increase an acting force applied by the elastic member **24** to the second clamping piece **212**. Therefore, specific space needs to be reserved on the side of the second clamping piece **212** that is away from the first clamping piece **211** along the first direction, so that the mounting base **23** can move toward the first clamping piece **211** relative to the second clamping piece **212**, and the side of the second clamping piece **212** that is away from the first clamping piece **211** along the first direction is definitely spaced away from the mounting base **23**. Because the force applying portion **231** is located on the side of the second clamping piece **212** that is away from the first clamping piece **211** along the first direction, the elastic member **24** is at least partially located on the side of the second clamping piece **212** that is away from the first clamping piece **211** along the first direction. In this way, space originally formed between the mounting base **23** and the side of the second clamping piece **212** that is away from the first clamping piece **211** along the first direction is fully utilized, so that space utilization is improved, and overall space occupied by the mounting base **23**, the second clamping piece **212**, and the elastic member **24** is reduced.

[0137] In an embodiment, as shown in FIG. 5, FIG. 6, FIG. 8, and FIG. 9, the mounting base **23** is further provided with a position limiting portion **232**, the position limiting portion **232** is located on a side of the force applying portion **231** that faces the first clamping piece **211** along the first direction, the second clamping piece **212** is located between the position limiting portion **232** and the force applying portion **231** along the first direction, and the second clamping piece **212** abuts against the position limiting portion **232** under the action of an elastic force of the elastic member **24**.

[0138] In this embodiment of this disclosure, the second clamping piece **212** is pressed against the position limiting portion **232** by the elastic force of the compressed elastic member **24**, so that shaking of the second clamping piece **212** between the position limiting portion **232** and the elastic member **24** is reduced.

[0139] In an embodiment, as shown in FIG. 5, FIG. 6, FIG. 8, and FIG. 9, the elastic member **24** is located on a side of the second clamping piece **212** that is away from the position limiting portion **232**, and the elastic member **24** is located between the second clamping piece **212** and the force applying portion **231**.

[0140] In this embodiment of this disclosure, two ends of the elastic member **24** respectively abut against the second clamping piece **212** and the force applying portion **231**. To be specific, an acting force applied by the force applying portion **231** can be transferred to the second clamping piece **212** through the elastic member **24**. This helps simplify a mounting structure of the elastic member **24**. The elastic member **24** is located between the second clamping piece **212** and the force applying

portion **231**. The elastic member **24** is neither located on a side of the second clamping piece **212** that is away from the force applying portion **231** nor located on a side of the force applying portion **231** that is away from the second clamping piece **212**. The elastic member **24** is fully located in space originally formed between the force applying portion **231** and a side of the second clamping piece **212** that is away from the first clamping piece **211**, and the space between the force applying portion **231** and the side of the second clamping piece **212** that is away from the first clamping piece **211** is fully utilized to accommodate the elastic member **24**, so that space utilization is improved.

[0141] In an embodiment, as shown in FIG. 4, FIG. 5, FIG. 7, and FIG. 8, each clamping assembly **200** has a plurality of clamping modules **2**, and clamping jaws **21** of the plurality of clamping modules **2** are arranged at spacings along the first direction.

[0142] In this embodiment of this disclosure, among the plurality of clamping modules **2** arranged at spacings along the first direction, each clamping module **2** correspondingly grabs a battery cell **600** on a conveying apparatus **400**, and the plurality of clamping modules **2** may jointly grab battery cells **600** on a plurality of conveying apparatuses **400**, so that efficiency of grabbing battery cells **600** is improved.

[0143] In an embodiment, as shown in FIG. 4, FIG. 5, FIG. 7, and FIG. 8, each clamping assembly **200** has two clamping modules **2**, two second clamping pieces **212** corresponding to the two clamping modules **2** are located between two first clamping pieces **211** along the first direction, two corresponding mounting bases **23** are located between the two first clamping pieces **211** along the first direction, and the second drive component **25** is located on a side of the mounting base **23** along the second direction.

[0144] In this embodiment of this disclosure, two corresponding mounting bases **23** are located between two first clamping pieces **211** along the first direction, the second drive component **25** is located on a side of the mounting base **23** along the second direction, and a spacing between the two mounting bases **23** at limiting positions away from the first clamping pieces **211** along the first direction can be arranged to be small, to help reduce an overall size of the grabbing device along the first direction.

[0145] In an embodiment, as shown in FIG. 5 and FIG. 6, a second drive component **25** of a clamping module **2** of each clamping assembly **200** is located on a side of a corresponding mounting base **23** that is away from another corresponding clamping assembly **200** along the second direction, and two second drive components **25** corresponding to two clamping modules **2** are located between two first clamping pieces **211** along the first direction.

[0146] In this embodiment of this disclosure, a second drive component **25** of a clamping module **2** of each clamping assembly **200** is located on a side of a corresponding mounting base **23** that is away from another corresponding clamping assembly **200** along the second direction, and the two clamping assemblies **200** can approach each other as close as possible along the second direction. This helps grab battery cells **600** at different spacings on the same conveying apparatus **400** within a larger spacing range.

[0147] In an embodiment, as shown in FIG. 6, a drive portion **251** is formed at an end of each second drive component **25** that faces a corresponding first clamping piece **211**, the drive portion **251** is connected to an end of a corresponding mounting base **23** that faces a corresponding first clamping piece **211**, and two second drive components **25** are located between two drive portions **251** along the first direction.

[0148] The second drive component **25** drives the mounting base **23** to move along the first direction. A size of the second drive component **25** along the first direction is large.

[0149] In this embodiment of this disclosure, the two second drive components **25** are located between the two drive portions **251** along the first direction, and the size of the second drive component **25** along the first direction is large, so that a spacing between the two drive portions **251** along the first direction is large, and a region between the two drive portions **251** spaced apart

along the first direction is large. The mounting base **23** moves relative to the second clamping piece **212**, and a size of the mounting base **23** along the first direction is large. Because the drive portion **251** is connected to an end of a corresponding mounting base **23** that faces a corresponding first clamping piece **211**, two mounting bases **23** with large sizes along the first direction can be arranged in a region between two drive portions **251** along the first direction, so that the region between the two drive portions **251** spaced apart along the first direction can be fully utilized, to improve space utilization.

[0150] In an embodiment, as shown in FIG. 5, FIG. 6, and FIG. 8, the clamping assembly **200** further includes a stop mechanism **3**, the stop mechanism **3** is located between two mounting bases **23** along the first direction, and the stop mechanism **3** is able to abut against the two mounting bases **23** along the first direction to prevent the two mounting bases **23** from approaching each other.

[0151] In this embodiment of this disclosure, the stop mechanism **3** between the two mounting bases **23** abuts against the two mounting bases **23** along the first direction to prevent the two mounting bases **23** from approaching each other, so that a spacing between the two mounting bases **23** abutting against the stop mechanism **3** reaches a minimum spacing that is possible between the two mounting bases **23**, and a possibility of interference between the two mounting bases **23** due to approaching is reduced.

[0152] In an embodiment, as shown in FIG. 10, the stop mechanism **3** includes a supporting base **31**, a stop rod **32**, and a locking piece **33**. The supporting base **31** is connected to the base **1**. The stop rod **32** passes through the supporting base **31**. Each mounting base **23** is correspondingly provided with the stop rod **32**, and the stop rod **32** is configured to abut against a corresponding mounting base **23**. The locking piece **33** is in threaded connection to a corresponding stop rod **32**. Each stop rod **32** is correspondingly provided with two locking pieces **33**. One locking piece **33** is located on a side of the supporting base **31** along the first direction, and the other locking piece **33** is located on the other side of the supporting base **31** along the first direction.

[0153] For example, the locking piece **33** is a nut.

[0154] In this embodiment of this disclosure, a stop rod **32** corresponding to each mounting base **23** passes through the supporting base **31**, and two side locking pieces **33** corresponding to each stop rod **32** are in threaded connection to the stop rod **32**, so that the stop rod **32** is fastened to the supporting base **31**. When one stop rod **32** is damaged, the corresponding stop rod **32** may be removed and replaced with a new stop rod **32**, without replacing the entire stop mechanism **3**. This helps reduce maintenance costs.

[0155] In an embodiment, as shown in FIG. 4, FIG. 5, and FIG. 7, a base **1** of one clamping assembly **200** is connected to the frame **100**, and a base **1** of another clamping assembly **200** is movably arranged on the frame **100**.

[0156] For example, as shown in FIG. 7, a base **1** of a clamping assembly **200** on the left in the figure is movably arranged on the frame **100**, and a base **1** of a clamping assembly **200** on the right in the figure is connected to the frame **100**.

[0157] In this embodiment of this disclosure, because a base **1** of one clamping assembly **200** is connected to the frame **100**, the corresponding clamping assembly **200** does not move relative to the base **1**. This helps simplify an assembly structure between the corresponding clamping assembly **200** and the base **1**. One of two clamping assemblies **200** moves relative to the base **1**, and the other clamping assembly **200** does not move relative to the base **1**, so that one clamping assembly **200** can move relative to the other clamping assembly **200**, and an assembly structure between the clamping assemblies **200** and the base **1** can also be simplified.

[0158] In an embodiment, as shown in FIG. 5 and FIG. 8, the base **1** includes a main base **11** and a position limiting base **12**. The second drive component **25**, the first clamping piece **211**, and the mounting base **23** are all arranged on the main base **11**. The position limiting base **12** is connected to the main base **11**. Both the position limiting base **12** and the first drive component **22** are located

on a side of the first clamping piece **211** that is away from the second clamping piece **212** along the first direction. The first drive component **22** is mounted on the position limiting base **12**, and the first drive component **22** is able to drive the first clamping piece **211** to move to abut against the position limiting base **12**.

[0159] In this embodiment of this disclosure, the position limiting base **12** is located at an end of the first clamping piece **211** that is away from the second clamping piece **212**, and the position limiting base **12** can abut against the first clamping piece **211** for position limiting, to prevent the first clamping piece **211** from excessively moving away from the second clamping piece **212** by a long distance. The first drive component **22** is mounted on the position limiting base **12** located on the side of the first clamping piece **211** that is away from the second clamping piece **212**. This facilitates mounting of the first drive component **22**. In addition, the first drive component **22** mounted on the position limiting base **12** is located on the side of the first clamping piece **211** that is away from the second clamping piece **212**, so that the first drive component **22** can smoothly push the first clamping piece **211** to move along the first direction.

[0160] An embodiment of this disclosure provides a clamping method. As shown in FIG. **11**, the clamping method includes the following steps.

[0161] Step **S101**: Drive, by using a first drive component, a first clamping piece to move to get in contact with a battery cell.

[0162] Step **S102**: Drive, by using a second drive component, a mounting base to move, so that a second clamping piece moves to get in contact with the battery cell.

[0163] Step **S103**: When the battery cell located between the first clamping piece and the second clamping piece is in contact with the first clamping piece and the second clamping piece, stop driving, by using the first drive component, the first clamping piece to move.

[0164] Step **S104**: Drive, by using the second drive component, the mounting base to move to a preset clamping position toward the first clamping piece that stops moving.

[0165] For example, step **S101** and step **S102** may be simultaneously performed to clamp the battery cell in a centered manner, and the first clamping piece and the second clamping piece get in contact with the battery cell at approximately the same time.

[0166] For example, step **S101** may be performed before step **S102**.

[0167] For example, step **S102** may be performed before step **S101**.

[0168] In this embodiment of this disclosure, when the battery cell **600** is in contact with the first clamping piece **211** and the second clamping piece **212**, the battery cell **600** is in a pre-clamped state, and the first clamping piece **211** and the second clamping piece **212** have not completely clamped the battery cell **600**. When the battery cell **600** is in contact with the first clamping piece **211** and the second clamping piece **212**, the first drive component **22** stops driving the first clamping piece **211** to move, to prevent the first clamping piece **211** from being pushed by the first drive component **22** to press against the battery cell **600**. When the battery cell **600** is in contact with the first clamping piece **211** and the second clamping piece **212**, the second drive component **25** drives the mounting base **23** to move to the preset clamping position toward the first clamping piece **211** that stops moving. As the mounting base **23** moves toward the first clamping piece **211**, deformation of an elastic member **24** gradually increases, so that a clamping force applied by a force applying portion **231** of the mounting base **23** to the second clamping piece **212** through the elastic member **24** and a reaction force applied by the first clamping piece **211** to the battery cell **600** increase. To be specific, a clamping force applied by the first clamping piece **211** and the second clamping piece **212** to clamp the battery cell **600** gradually increases. This helps reduce a possibility that the first clamping piece **211** and the second clamping piece **212** cause damage to the battery cell **600** in a process of clamping the battery cell **600**. In addition, no deformable buffer component is arranged between the first clamping piece **211** and the first drive component **22**, and a position to which the first drive component **22** performs driving determines a position of the first clamping piece **211**. When the first drive component **22** stops driving the first clamping piece **211**

to move, the position of the first clamping piece **211** no longer changes, so that the first clamping piece **211** can well limit a position of the clamped battery cell **600**. This can reduce shaking of the clamped battery cell **600**, and help stably clamp the battery cell **600**.

[0169] An embodiment of this disclosure provides a transfer method. As shown in FIG. **12**, the method includes the following steps.

[0170] Step **S1**: Drive, by using a transfer device, a grabbing device to move to a to-be-grabbed material position on a conveying apparatus, so that a battery cell is located between a first clamping piece and a corresponding second clamping piece.

[0171] Step **S2**: Drive, by using a first drive component, the first clamping piece to move to get in contact with the battery cell.

[0172] Step **S3**: Drive, by using a second drive component, a mounting base to move, so that the second clamping piece moves to get in contact with the battery cell.

[0173] Step **S4**: When the battery cell located between the first clamping piece and the second clamping piece is in contact with the first clamping piece and the second clamping piece, stop driving, by using the first drive component, the first clamping piece to move.

[0174] Step **S5**: Drive, by using the second drive component, the mounting base to move, so that the second clamping piece moves to a preset clamping position toward the first clamping piece that stops moving.

[0175] Step **S6**: Transfer, by using the transfer device, the battery cell grabbed by the grabbing device to a stacking table for stacking.

[0176] For example, FIG. **2** shows a state in which single-row-assembled battery cells **600** are stacked on the stacking table **500**.

[0177] For example, FIG. **3** shows a state in which double-row-assembled battery cells **600** are stacked on the stacking table **500**.

[0178] In this embodiment of this disclosure, the transfer device drives the grabbing device to move to the to-be-grabbed material position, to prepare to grab the battery cell **600** on the conveying apparatus **400**. The first drive component **22** and the second drive component **25** independently perform driving, so that the first clamping piece **211** and the second clamping piece **212** can stably clamp the battery cell **600** while reducing a possibility of causing damage to the clamped battery cell **600**. Therefore, the grabbing device can stably grab the battery cell **600** on the conveying apparatus **400** while reducing damage to the battery cell **600**, and transfer the grabbed battery cell **600** to the stacking table **500** for stacking. In this way, the battery cell **600** on the conveying apparatus **400** can be well transferred to the stacking table **500**.

[0179] In an embodiment, as shown in FIG. **13**, the transfer method further includes the following step:

[0180] Step **S7**: Drive, by using a drive assembly, one of two clamping assemblies to move relative to the other clamping assembly, so that a spacing between two clamping jaws arranged along a second direction is suitable for grabbing assembled battery cells on the conveying apparatus.

[0181] The assembled battery cells **600** may have a variety of specifications. For assembled battery cells **600** with different specifications, a spacing required between the two clamping jaws **21** may vary.

[0182] For example, the assembled battery cells **600** may be single-row-assembled battery cells **600** or double-row-assembled battery cells **600**. A spacing required between the two clamping jaws **21** arranged along the second direction for single-row-assembled battery cells **600** is a first spacing. A spacing required between the two clamping jaws **21** arranged along the second direction for double-row-assembled battery cells **600** is a second spacing. The first spacing and the second spacing are unequal.

[0183] For example, two groups of single-row-assembled battery cells **600** with different specifications may require different spacings between the two clamping jaws **21**, or two groups of double-row-assembled battery cells **600** with different specifications may require different spacings

between the two clamping jaws **21**. For example, a spacing required between the two clamping jaws **21** for one group of single-row-assembled battery cells **600** is a third spacing, and a spacing required between the two clamping jaws **21** for the other group of single-row-assembled battery cells **600** is a third spacing, where the third spacing and the fourth spacing are unequal.

[0184] In this embodiment of this disclosure, the drive assembly **300** drives one clamping assembly **200** to move relative to another clamping assembly **200**, so that a spacing between two clamping modules **2** arranged along the second direction can be changed, and a spacing between two clamping jaws **21** arranged along the second direction is adjusted. In this way, the spacing between the two clamping jaws **21** arranged along the second direction can be suitable for respectively grabbing two assembled battery cells **600** on the conveying apparatus **400**.

[0185] In an embodiment, as shown in FIG. **8**, there are two conveying apparatuses **400**, the two conveying apparatuses **400** are arranged in parallel, each clamping assembly **200** has two clamping modules **2**, the two clamping modules **2** of each clamping assembly **200** are in a one-to-one correspondence to the conveying apparatuses **400**, and a pole of a battery cell **600** on each conveying apparatus **400** faces the other corresponding conveying apparatus **400**.

[0186] It should be noted that the two conveying apparatuses **400** are arranged in parallel, and an arrangement direction of the two conveying apparatuses **400** intersects with a conveying direction of the conveying apparatuses **400**.

[0187] For example, the arrangement direction of the two conveying apparatuses **400** is perpendicular to the conveying direction.

[0188] Two clamping modules **2** of each clamping assembly **200** are in a one-to-one correspondence to the conveying apparatuses **400**. One of two clamping modules **2** of each clamping assembly **200** is configured to grab a battery cell **600** on one conveying apparatus **400**, and the other clamping module **2** is configured to grab a battery cell **600** on the other conveying apparatus **400**.

[0189] In this embodiment of this disclosure, a pole of a battery cell **600** on each conveying apparatus **400** faces the other corresponding conveying apparatus **400**. Therefore, when two clamping modules **2** of each clamping assembly **200** respectively grab battery cells **600** on the two conveying apparatuses **400**, a pole of a battery cell **600** clamped by each of the two clamping modules **2** faces the other clamping module **2**. When poles of battery cells **600** clamped by two clamping modules **2** of each clamping assembly **200** are arranged along a vertical direction, because a pole of a battery cell **600** clamped by each of the two clamping modules **2** faces the other corresponding clamping module **2**, a pole of a battery cell **600** clamped by one clamping module **2** inevitably faces upward. The grabbing device rotates, so that positions of two clamping modules **2** of each clamping assembly **200** along the vertical direction can be interchanged, and a pole of a battery cell **600** clamped by the other clamping module **2** can be adjusted to face upward.

[0190] In an embodiment, as shown in FIG. **13**, the transferring, by using the transfer device, the battery cell grabbed by the grabbing device to a stacking table for stacking includes the following steps:

[0191] Step **S61**: Drive, by using the transfer device, the grabbing device to move away from the conveying apparatus, so that the grabbing device is located at a position able to prevent the grabbing device from interfering with the conveying apparatus during rotation.

[0192] Step **S62**: Drive, by using the transfer device, the grabbing device to rotate, so that a pole of the battery cell grabbed by the grabbing device is arranged along a vertical direction.

[0193] Step **S63**: Perform stacking: Drive, by using the transfer device, the grabbing device to move, so that the grabbing device transfers the grabbed battery cell whose pole faces upward to the stacking table for stacking and release.

[0194] Step **S64**: When battery cells grabbed by the grabbing device are not all released, drive, by using the transfer device, the grabbing device that has released a battery cell to rotate in an avoidance region, so that a pole of a battery cell grabbed by the grabbing device faces upward for

performing the stacking step.

[0195] The avoidance region is mainly used for avoiding the stacking table **500**. The grabbing device rotates in the avoidance region, so that interference between the grabbing device and the stacking table **500** can be avoided.

[0196] In this embodiment of this disclosure, the transfer device drives the grabbing device to move away from the conveying apparatus **400**, to prevent the grabbing device from interfering with the conveying apparatus **400** during rotation. Because a pole of a battery cell **600** on each conveying apparatus **400** faces the other corresponding conveying apparatus **400**, poles of battery cells **600** on the conveying apparatuses **400** are approximately arranged along a horizontal direction. However, a pole of a battery cell **600** to be stacked on the stacking table **500** needs to face upward. The transfer device drives the grabbing device to rotate, so that a pole of a battery cell **600** grabbed by the grabbing device is arranged along the vertical direction. Because poles of battery cells **600** clamped by two clamping modules **2** of each clamping assembly **200** face different directions, a pole of at least one of two battery cells **600** arranged along the vertical direction faces upward. The transfer device may drive the grabbing device to move, to transfer a battery cell **600** whose pole faces upward to the stacking table **500** for stacking, and a corresponding clamping jaw **21** may be released to release a stacked battery cell **600**. A battery cell **600** clamped by one of two clamping modules **2** of each clamping assembly **200** is stacked and released, and the other clamping module **2** is still clamping a battery cell **600**. Battery cells **600** grabbed by the grabbing device are not all released. The transfer device drives the grabbing device that has released a battery cell **600** to rotate in the avoidance region, so that a pole of a battery cell **600** grabbed by the grabbing device faces upward. Through rotation of the grabbing device, a pole of a battery cell **600** that has not been released by the grabbing device switches from downward arrangement to upward arrangement. Then a stacking step is performed again, so that a battery cell **600** whose pole faces upward moves to the stacking table **500** for stacking and release. In this way, transfer and release of all battery cells **600** on the grabbing device are completed.

[0197] In an embodiment, as shown in FIG. **13**, the transferring, by using the transfer device, the battery cell grabbed by the grabbing device to a stacking table for stacking further includes the following step:

[0198] Step **S65**: When the battery cell grabbed by the grabbing device is single-row-assembled battery cells, drive, by using the drive assembly, one of two clamping assemblies to move relative to the other clamping assembly, so that a spacing between two clamping jaws arranged along the second direction is suitable for placing the single-row-assembled battery cells to a corresponding position on the stacking table.

[0199] In this embodiment of this disclosure, for single-row-assembled battery cells **600**, a spacing between two single-row-assembled battery cells **600** on the conveying apparatus **400** is different from a spacing between the two single-row-assembled battery cells **600** on the stacking table **500**. In a process in which the grabbing device grabs single-row-assembled battery cells **600** from the conveying apparatus **400** and transfers the battery cells **600** to the stacking table **500**, the drive assembly **300** drives one of two clamping assemblies **200** to move relative to the other clamping assembly **200**, to adjust a spacing between two clamping jaws **21** arranged along the second direction. Correspondingly, a spacing between battery cells **600** clamped by the two clamping jaws **21** arranged along the second direction is changed, so that the spacing between the battery cells **600** clamped by the two clamping jaws **21** arranged along the second direction can match a position for placing two single-row-assembled battery cells **600** on the stacking table **500**.

[0200] As shown in FIG. **14**, a specific process of transferring a battery cell **600** by using the transfer method in the embodiments of this disclosure is as follows.

[0201] Step **S301**: Drive, by using a transfer device, a grabbing device to move from an initial position to a position above a conveying apparatus.

[0202] Step **S302**: Drive, by using a drive assembly, one clamping assembly to move relative to

another clamping assembly, so that a spacing between two clamping jaws arranged along a second direction is suitable for grabbing assembled battery cells on the conveying apparatus.

[0203] Step **S303**: Drive, by using the transfer device, the grabbing device to move, so that the two clamping jaws arranged along the second direction are respectively located above two assembled battery cells on the conveying apparatus.

[0204] Step **S304**: Drive, by using the transfer device, the grabbing device to move to a to-be-grabbed material position on the conveying apparatus, so that the two assembled battery cells are located between a corresponding first clamping piece and a corresponding second clamping piece.

[0205] Step **S305**: Drive, by using a first drive component, the first clamping piece to move to get in contact with a corresponding battery cell.

[0206] Step **S306**: Drive, by using a second drive component, a mounting base to move, so that the second clamping piece moves to get in contact with the battery cell.

[0207] Step **S307**: When the battery cell located between the first clamping piece and the second clamping piece is in contact with the first clamping piece and the second clamping piece, stop driving, by using the first drive component, the first clamping piece to move.

[0208] Step **S308**: Drive, by using the second drive component, the mounting base to move to a preset clamping position toward the first clamping piece that stops moving.

[0209] Step **S309**: Drive, by using the transfer device, the grabbing device to move away from the conveying apparatus, so that the grabbing device is located at a position able to prevent the grabbing device from interfering with the conveying apparatus during rotation.

[0210] Step **S310**: Drive, by using the transfer device, the grabbing device to rotate, so that a pole of the battery cell grabbed by the grabbing device is arranged along a vertical direction.

[0211] Step **S311**: Drive, by using the transfer device, the grabbing device to move, so that the grabbing device transfers the grabbed battery cell whose pole faces upward to the stacking table for stacking and release.

[0212] Step **S312**: Drive, by using the transfer device, the grabbing device to move to an avoidance region.

[0213] Step **S313**: Drive, by using the transfer device, the grabbing device to rotate in the avoidance region, so that the grabbing device drives a battery cell whose pole faces downward to move until the pole faces upward.

[0214] Step **S314**: Drive, by using the transfer device, the grabbing device to move, so that the grabbing device transfers the grabbed battery cell whose pole faces upward to the stacking table for stacking and release.

[0215] Step **S315**: Drive, by using the transfer device, the grabbing device to return from the stacking table to the initial position to wait for next grabbing.

[0216] The foregoing embodiments are merely intended to describe the technical solutions of this disclosure, but not to limit this disclosure. Although this disclosure is described in detail with reference to the foregoing embodiments, persons of ordinary skill in the art should understand that they may still make modifications to the technical solutions described in the foregoing embodiments or make equivalent replacements to some or all technical features thereof, without departing from the scope of the technical solutions in the embodiments of this disclosure. All such modifications and equivalent replacements shall fall within the scope of this disclosure. In particular, as long as there is no structural conflict, the various technical features mentioned in the embodiments can be combined in any manner.

INDUSTRIAL PRACTICABILITY

[0217] In a clamping module in the embodiments of this disclosure, before a clamping jaw clamps a battery cell, the battery cell is located between a first clamping piece and a second clamping piece along a first direction, and the battery cell is spaced away from the first clamping piece and the second clamping piece by a specific distance along the first direction. In this way, in a process in which the clamping jaw moves downward to a position at which the battery cell can be grabbed, a

possibility that the first clamping piece and the second clamping piece press against the battery cell and cause damage to the battery cell can be reduced. Because the battery cell is spaced away from the first clamping piece and the second clamping piece by a specific distance along the first direction, a first drive component needs to drive the first clamping piece to move along the first direction to a position for contact with the battery cell, and a second drive component needs to drive a mounting base to move along the first direction to enable the second clamping piece to move to a position for contact with the battery cell. When the battery cell is in contact with the first clamping piece and the second clamping piece, the first drive component stops driving the first clamping piece to move along the first direction, and the second drive component drives the mounting base to move toward the first clamping piece, so that a force applying portion of the mounting base gradually increases, through an elastic member, an acting force applied to the second clamping piece. A clamping force applied by the first clamping piece and the second clamping piece to clamp the battery cell is gradually increased to a required clamping force through buffering and adjustment by the elastic member, to reduce a possibility that the clamping force applied by the first clamping piece and the second clamping piece to the battery cell is excessively large and causes damage to the battery cell. In addition, the second drive component drives the mounting base to move to gradually increase the clamping force, so that the battery cell can be stably clamped. In a process in which the second drive component drives the mounting base to move toward the first clamping piece to enable the force applying portion of the mounting base to gradually increase, through the elastic member, the acting force applied to the second clamping piece, because the first drive component drives the first clamping piece to move and no corresponding elastic member is arranged between the first drive component and the first clamping piece, the first drive component rigidly drives the first clamping piece, and a position to which the first drive component performs driving determines a position of the first clamping piece. This can reduce shaking of the battery cell clamped between the first clamping piece and the second clamping piece, and help stably clamp the battery cell.

Claims

1. A clamping module, comprising: a clamping jaw, comprising a first clamping piece and a second clamping piece that are arranged opposite each other, wherein a direction in which the first clamping piece and the second clamping piece are arranged opposite each other is a first direction; a first drive component, configured to drive the first clamping piece to move along the first direction; a mounting base arranged on the second clamping piece, wherein the second clamping piece is able to move relative to the mounting base along the first direction, and the mounting base is provided with a force applying portion that is spaced away from the second clamping piece; an elastic member in contact with the second clamping piece and the force applying portion, to enable the force applying portion to apply an acting force to the second clamping piece through the elastic member; and a second drive component, configured to drive the mounting base to move along the first direction, so that a direction of the acting force applied by the elastic member to the second clamping piece is toward the first clamping piece.
2. The clamping module according to claim 1, wherein the force applying portion is located on a side of the second clamping piece that is away from the first clamping piece along the first direction.
3. The clamping module according to claim 2, wherein the mounting base is further provided with a position limiting portion, the position limiting portion is located on a side of the force applying portion that faces the first clamping piece along the first direction, the second clamping piece is located between the position limiting portion and the force applying portion along the first direction, and the second clamping piece abuts against the position limiting portion under the action of an elastic force of the elastic member.

4. The clamping module according to claim 3, wherein the elastic member is located on a side of the second clamping piece that is away from the position limiting portion, and the elastic member is located between the second clamping piece and the force applying portion.
5. A grabbing device, comprising: a frame; clamping assemblies arranged on the frame, wherein there are at least two clamping assemblies, each clamping assembly comprises a base and the clamping module according to claim 1, at least two clamping assemblies are arranged along a second direction, the second direction intersects with the first direction, the first drive component, the second drive component, the first clamping piece, and the mounting base are all arranged on the base, the first drive component drives the first clamping piece to move relative to the base, the second drive component drives the mounting base to move relative to the base, each clamping assembly has at least one clamping module, and each clamping module of each clamping assembly and a corresponding clamping module of another corresponding clamping assembly are arranged along the second direction; and a drive assembly arranged on the frame, wherein the drive assembly is configured to drive one clamping assembly to move relative to another clamping assembly along the second direction.
6. The grabbing device according to claim 5, wherein each clamping assembly has a plurality of clamping modules, and clamping jaws of the plurality of clamping modules are arranged at spacings along the first direction.
7. The grabbing device according to claim 6, wherein each clamping assembly has two clamping modules, two second clamping pieces corresponding to the two clamping modules are located between two first clamping pieces along the first direction, two corresponding mounting bases are located between the two first clamping pieces along the first direction, and the second drive component is located on a side of the mounting base along the second direction.
8. The grabbing device according to claim 7, wherein a second drive component of a clamping module of each clamping assembly is located on a side of a corresponding mounting base that is away from another corresponding clamping assembly along the second direction, and two second drive components corresponding to two clamping modules are located between two first clamping pieces along the first direction.
9. The grabbing device according to claim 8, wherein a drive portion is formed at an end of each second drive component that faces a corresponding first clamping piece, the drive portion is connected to an end of a corresponding mounting base that faces a corresponding first clamping piece, and two second drive components are located between two drive portions along the first direction.
10. The grabbing device according to claim 7, wherein the clamping assembly further comprises a stop mechanism, the stop mechanism is located between two mounting bases along the first direction, and the stop mechanism is able to abut against the two mounting bases along the first direction to prevent the two mounting bases from approaching each other.
11. The grabbing device according to claim 10, wherein the stop mechanism comprises: a supporting base connected to the base; a stop rod passing through the supporting base, wherein each mounting base is correspondingly provided with the stop rod, and the stop rod is configured to abut against a corresponding mounting base; and a locking piece in threaded connection to a corresponding stop rod, wherein each stop rod is correspondingly provided with two locking pieces, one locking piece is located on a side of the supporting base along the first direction, and the other locking piece is located on the other side of the supporting base along the first direction.
12. The grabbing device according to claim 5, wherein a base of one clamping assembly is connected to the frame, and a base of another clamping assembly is movably arranged on the frame.
13. The grabbing device according to claim 5, wherein the base comprises: a main base, wherein the second drive component, the first clamping piece, and the mounting base are all arranged on the main base; and a position limiting base connected to the main base, wherein both the position

limiting base and the first drive component are located on a side of the first clamping piece that is away from the second clamping piece along the first direction, the first drive component is mounted on the position limiting base, and the first drive component is able to drive the first clamping piece to move to abut against the position limiting base.

14. A battery production line, comprising: a conveying apparatus, configured to convey a battery cell; the grabbing device according to claim 5, configured to grab the battery cell conveyed by the conveying apparatus; a stacking table, configured to receive the battery cell grabbed by the grabbing device from the conveying apparatus, to pre-stack the battery cell; and a transfer device, configured to drive the grabbing device to reciprocate between the conveying apparatus and the stacking table.

15. A clamping method, comprising: driving, by using a first drive component, a first clamping piece to move to get in contact with a battery cell; driving, by using a second drive component, a mounting base to move, so that a second clamping piece moves to get in contact with the battery cell; when the battery cell located between the first clamping piece and the second clamping piece is in contact with the first clamping piece and the second clamping piece, stopping driving, by using the first drive component, the first clamping piece to move; and driving, by using the second drive component, the mounting base to move to a preset clamping position toward the first clamping piece that stops moving.

16. A transfer method, comprising: driving, by using a transfer device, a grabbing device to move to a to-be-grabbed material position on a conveying apparatus, so that a battery cell is located between a first clamping piece and a corresponding second clamping piece; driving, by using a first drive component, the first clamping piece to move to get in contact with the battery cell; driving, by using a second drive component, a mounting base to move, so that the second clamping piece moves to get in contact with the battery cell; when the battery cell located between the first clamping piece and the second clamping piece is in contact with the first clamping piece and the second clamping piece, stopping driving, by using the first drive component, the first clamping piece to move; driving, by using the second drive component, the mounting base to move, so that the second clamping piece moves to a preset clamping position toward the first clamping piece that stops moving; and transferring, by using the transfer device, the battery cell grabbed by the grabbing device to a stacking table for stacking.

17. The transfer method according to claim 16, wherein the transfer method further comprises: driving, by using a drive assembly, one of two clamping assemblies to move relative to the other clamping assembly, so that a spacing between two clamping jaws arranged along a second direction is suitable for grabbing assembled battery cells on the conveying apparatus.

18. The transfer method according to claim 16, wherein there are two conveying apparatuses, the two conveying apparatuses are arranged in parallel, each clamping assembly has two clamping modules, the two clamping modules of each clamping assembly are in a one-to-one correspondence to the conveying apparatuses, and a pole of a battery cell on each conveying apparatus faces the other corresponding conveying apparatus; and the transferring, by using the transfer device, the battery cell grabbed by the grabbing device to a stacking table for stacking comprises: driving, by using the transfer device, the grabbing device to move away from the conveying apparatus, so that the grabbing device is located at a position able to prevent the grabbing device from interfering with the conveying apparatus during rotation; driving, by using the transfer device, the grabbing device to rotate, so that a pole of the battery cell grabbed by the grabbing device is arranged along a vertical direction; performing stacking: driving, by using the transfer device, the grabbing device to move, so that the grabbing device transfers the grabbed battery cell whose pole faces upward to the stacking table for stacking and release; and when battery cells grabbed by the grabbing device are not all released, driving, by using the transfer device, the grabbing device that has released a battery cell to rotate in an avoidance region, so that a pole of a battery cell grabbed by the grabbing device faces upward for performing the stacking step.

19. The transfer method according to claim 18, wherein the transferring, by using the transfer device, the battery cell grabbed by the grabbing device to a stacking table for stacking further comprises: when the battery cell grabbed by the grabbing device is single-row-assembled battery cells, driving, by using the drive assembly, one of two clamping assemblies to move relative to the other clamping assembly, so that a spacing between two clamping jaws arranged along the second direction is suitable for placing the single-row-assembled battery cells to a corresponding position on the stacking table.
